Diagnosing and Repairing Conservation-Law Failures in Physics-Informed Neural Networks for Liouville Transport: From Chaotic Classical Dynamics to Quantum Thermodynamics

[Author name – fill in][1]

[1] *[Affiliation – fill in]*

(Dated: August 13, 2026)

Split-operator techniques (SOT) based on fast Fourier transforms are an established, accurate method for propagating classical phase-space distribution functions under the Liouville equation, including in chaotic regimes such as periodically δ-kicked double-well oscillators [Gómez, Thirumuruganandham, and Santana, Comput. Phys. Commun. **185**, 136 (2014)]. Physics-informed neural networks (PINNs) have been proposed as mesh-free alternatives, motivated by the curse of dimensionality that limits SOT to low-dimensional phase spaces. This paper's contribution is not the observation that a PINN can fail, nor the use of a Hamiltonian neural network (HNN) as a fix – both architectures are established. The contribution is threefold: a specific, previously unisolated mechanism for *why* residual-only PINNs fail on Liouville-type transport; independent confirmation of that mechanism on a quantum system where the analogous failure has been separately reported; and a mathematical, not merely empirical, account of why a structurally-constrained alternative resolves it, together with an explicit test of where that resolution's advantage holds and where it erodes.

We first show that a standard, residual-only PINN fails on the smooth Hamiltonian flow underlying the double-well system above: it violates total-probability conservation by a factor of 3.6 and ensemble-averaged energy conservation by a factor of order 270, despite a low training loss. Time-resolved analysis and a set of ablations (activation function, collocation density, initial-condition weighting) localize the defect to under-constraint of the network far from the initial condition's support – a low-amplitude background that trivially satisfies the residual almost everywhere in an oversized training domain – not to chaos or long-horizon error accumulation. An independent test on a driven-dissipative qubit (a Lindblad master equation, the quantum generalization of the Liouville operator underlying quantum thermodynamic relaxation and decoherence), robust across a sevenfold range of dissipation rates, supports this diagnosis: the same residual-only philosophy conserves trace to within 0.65% in that lower-dimensional, densely-sampled setting, two to three orders of magnitude better than the classical case, consistent with an independent report of the same qualitative failure in quantum dissipative dynamics [Ullah, Huang, Yang, and Dral, Digital Discovery **3**, 2052 (2024)].

Guided by this diagnosis, we implement and test two repairs, of increasing structural depth, rather than merely proposing them. The first, a hard-constraint reparametrization enforcing mass conservation and positivity by construction combined with sparse supervision from our own exact characteristics-based solver, reduces pointwise error by a factor of 8 and recovers the correct sign of the phase-space shearing that the original network missed entirely. The second replaces density-field regression with a Hamiltonian neural network [16] trained only on local equations-of-motion data, with the density then propagated through its learned dynamics via the same exact method of characteristics used for our reference solution. This conserves mass to 1.0000042 automatically, for a provable reason: any Hamiltonian vector field is exactly divergence-free by equality of mixed partial derivatives, independent of how accurately the learned generating function approximates the true one. It also achieves the lowest pointwise error of any configuration tested, an 11× improvement over the original failing configuration.

We then test this second repair's actual limits, rather than stopping at its best case. Extended to the real chaotic system that motivated this study – the full periodically δ-kicked double well – it correctly reproduces the qualitative chaotic folding structure through four kick periods, while pointwise error grows roughly 15× over that range, consistent with chaotic amplification of small errors at a rate set by the system's Lyapunov exponent. Repeating the chaos test across five random seeds reveals a further, previously unreported caveat: chaos-tracking accuracy varies by up to 38× depending on initialization alone, and final training loss is a poor predictor of which seed will perform best. An initial extension to a small network of two coupled oscillators shows a third, specific limit: the method learns the linear part of the coupled dynamics almost exactly but the nonlinear inter-node coupling term markedly less well, localizing where this approach's difficulty concentrates as complexity increases rather than leaving it diffuse. We report the failure, its mechanism, both repairs, and all three boundary tests together, and are explicit that this combination of results, not a claim that either architecture is individually novel, is the paper's contribution.

I. INTRODUCTION

Split-operator techniques (SOT) propagate the classical Liouville equation

$$\frac{\partial \rho(q,p;t)}{\partial t} = \hat{L}\rho(q,p;t), \qquad \hat{L} = \frac{\partial H}{\partial q}\frac{\partial}{\partial p} - \frac{\partial H}{\partial p}\frac{\partial}{\partial q} \quad (1)$$

on a fixed phase-space grid using fast Fourier transforms, building on spectral time-propagation methods developed originally for the Schrödinger equation [20], and achieving $O(N \log N)$ scaling per time step for an N-point grid [1]. Reference [1] demonstrated this method's effectiveness across regular, chaotic, dissipative, and relativistic classical dynamics, including a periodically δ-kicked quartic double-well oscillator that exhibits chaotic phase-space mixing (their Fig. 2). The authors explicitly note the method's principal limitation: grid-based propagation scales unfavorably with phase-space dimension, since the number of grid points required grows as N^d for d phase-space dimensions.

Physics-informed neural networks (PINNs) [2] approximate PDE solutions with a neural network trained to minimize the PDE residual (via automatic differentiation) plus initial/boundary condition losses, without requiring a discretized grid. This mesh-free property is often motivated as a route around the curse of dimensionality that limits grid-based methods. It is therefore natural to ask whether a PINN could serve as an alternative to SOT for the same class of chaotic classical systems studied in Ref. [1], particularly at phase-space dimensions where SOT becomes impractical.

Before addressing higher-dimensional cases, a necessary first test is whether a PINN can reproduce SOT-quality results on the *same* low-dimensional chaotic system where SOT is already known to work well. This is the question we address here. We deliberately test a single period of the smooth (unkicked) Hamiltonian flow underlying Ref. [1]'s chaotic double well, prior to any stroboscopic kicks or filamentation, since if a PINN cannot reproduce this simpler sub-problem, testing the full chaotic map is not yet meaningful.

Convection-dominated and pure-transport PDEs are a documented difficulty for PINNs. Krishnapriyan *et al.* [3] showed that PINNs can fail on convection equations even when the PDE residual loss is successfully minimized, because low residual does not guarantee convergence to the physically correct solution when the loss landscape admits non-physical near-minimizers. The Liouville equation, Eq. (1), is exactly such a transport equation: it has no diffusive term to regularize sharp features, and its solution is a rigid advection of the initial density along Hamiltonian trajectories.

Neither the specific failure mode we identify, nor either repair, is contingent on any single component being individually novel – PINNs [2], Hamiltonian neural networks [16], and hard-constraint reparametrization [8, 19] are all established. This paper's contribution is the following, specific combination, which to our knowledge has not been previously reported together:

(i) a mechanistic diagnosis, not merely an empirical observation, of why residual-only PINNs violate conservation laws on Liouville-type transport: an under-constrained network settling into a low-amplitude background that trivially satisfies the PDE residual over most of an oversized training domain (Sec. V);

(ii) independent confirmation of this mechanism's dimensionality-dependence on a quantum system where an analogous failure has been separately reported [8], including a robustness check across the system's physical parameter range rather than a single tested value (Sec. VI);

(iii) a mathematical, not only empirical, account of why a structurally-constrained repair resolves the failure – a proof that any Hamiltonian-generated vector field is exactly divergence-free regardless of the generating function's accuracy (Sec. VIII) – together with a weaker, explicitly compared repair that enforces the same conservation law by numerical construction rather than architectural guarantee (Sec. VII); and

(iv) explicit tests of where the stronger repair's advantage holds and where it erodes: under the system's actual chaotic dynamics (Sec. VIII.D), across random initialization (Sec. VIII.E), and under an initial extension to a coupled multi-node system (Sec. IX) – reported together with the successes, not separately or omitted.

II. RELATED WORK

PINNs applied to Liouville-type transport equations are not unprecedented, and situating this work against that literature is essential both for assessing novelty and for interpreting our result correctly. The idea of training a neural network against a differential-equation residual dates to Lagaris, Likas, and Fotiadis [12], who also introduced the construction we return to in Sec. VII: writing the network's output as a term satisfying initial/boundary conditions exactly plus a free term, rather than penalizing violations in the loss. [4] give a broad review of physics-informed machine learning as a field, [5] survey PINN methodology and open challenges in detail, and Han, Jentzen, and E [13] demonstrated deep-learning-based solvers scaling to genuinely high-dimensional PDEs, the regime where SOT's $O(N^d)$ grid scaling becomes prohibitive and where a working PINN alternative would matter most; none of these are specific to transport or Liouville-class equations. Wang, Teng, and Perdikaris [14] and Wang, Yu, and Perdikaris [15] analyze PINN training pathologies through gradient-flow and neural-tangent-kernel lenses respectively, providing a more theoretical account of failure modes complementary to the empirical ablations in Sec. V.B.

Closest to our classical test, Zhang *et al.* [6] developed PINN-Vlasov, applying physics-informed neural networks to the Vlasov-Poisson equation – the collisionless Liouville equation for a self-consistent plasma – and reported success on both forward and inverse problems. A critical methodological difference from our work is that PINN-Vlasov's training data comprised distribution-function values sampled from a fully kinetic particle-in-cell simulation, in addition to the PDE residual; that is, the network was supervised with labeled trajectory data, not trained on the residual and initial condition alone as in our initial experiment. This is a plausible candidate explanation for the difference in outcomes, and Sec. VII tests it directly by adding analogous (but much cheaper) data supervision to our own problem.

On the quantum side, two papers are directly relevant. Zeng, Kou, and Sun [7] tested neural-network propagators, including for the quantum Liouville equation, and found that *linear* fully-connected networks without hidden layers achieve high accuracy while preserving population conservation, attributing this to the natural alignment between a linear network's structure and the inherent linearity of the quantum propagator. Most directly relevant of all, Ullah *et al.* [8] studied trace conservation – the quantum analog of the total-probability conservation we test classically – in neural-network simulations of quantum dissipative dynamics, and found independently that standard neural network approaches violate it, that physics-informed training mitigates but does not eliminate the violation, and proposed a hard-constraint, uncertainty-aware correction applied outside the loss function that enforces trace conservation exactly by construction rather than encouraging it softly. Our classical mass-conservation finding (Sec. V) is, to our knowledge, an independent instance of the same qualitative failure mode Ullah *et al.* report on the quantum side; Sec. VI includes a direct test of our own on an open quantum system to examine this connection further, and Sec. VII implements and tests the classical analog of their hard-constraint remedy.

Finally, architectures that build conservation laws into their structure rather than their loss function are an active area with direct bearing on Sec. VII: Hamiltonian neural networks [16] and Lagrangian neural networks [17] parametrize dynamics so that energy conservation holds by construction, symplectic recurrent networks [18] do the same for the symplectic structure of Hamiltonian flow specifically, and hard-constraint PINN formulations have recently been developed directly for PDE-constrained optimal control problems [19]. Our Sec. VII fix is a direct, if simpler, instance of this broader design philosophy applied to mass conservation in a transport equation.

III. MODEL SYSTEM AND EXACT REFERENCE SOLUTION

We consider the quartic double-well Hamiltonian from Ref. [1] (Sec. 3, their Eq. for H_0), with $m = \omega_0 = D = 1$:

$$H_0(q,p) = \frac{p^2}{2} - \frac{q^2}{2} + \frac{q^4}{100}. \tag{2}$$

The associated Liouville transport equation for the phase-space density is

$$\frac{\partial \rho}{\partial t} + p\,\frac{\partial \rho}{\partial q} - V'(q)\,\frac{\partial \rho}{\partial p} = 0, \qquad V'(q) = -q + \frac{4q^3}{100}. \tag{3}$$

We restrict attention to the smooth Hamiltonian flow over a single period $t \in [0,1]$, i.e., prior to any δ-kick, so that any discrepancy we observe cannot be attributed to the kick discontinuity or to chaotic filamentation accumulated over many periods.

Because Eq. (2) is non-dissipative, Liouville's theorem gives an exact solution by the method of characteristics:

$$\rho(q,p;t) = \rho_0\big(\Phi_{-t}(q,p)\big), \tag{4}$$

where Φ_t is the Hamiltonian flow map generated by Eqs. of motion $\dot{q} = p$, $\dot{p} = -V'(q)$. We compute this reference by integrating Hamilton's equations backward in time from every evaluation point using fourth-order Runge-Kutta (200 substeps), which we verified conserves total probability mass to within 7×10^{-15} of unity on a 128×128 grid over $q, p \in [-15, 15]$. This serves as ground truth without requiring an independent from-scratch reimplementation of the SOT method to be blindly trusted.

The initial density is a normalized Gaussian,

$$\rho_0(q,p) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(q-q_0)^2 + (p-p_0)^2}{2\sigma^2}\right], \tag{5}$$

with $(q_0, p_0) = (1, 0)$ and $\sigma = 1$.

IV. PINN CONSTRUCTION

We train a fully-connected network $\rho_\theta(q,p,t)$ with four hidden layers of 64 units and tanh activations, taking (q,p,t) as input and a scalar density as output. The loss function is

$$\mathcal{L} = \mathcal{L}_{\mathrm{PDE}} + \lambda\,\mathcal{L}_{\mathrm{IC}}, \tag{6}$$

where $\mathcal{L}_{\mathrm{PDE}}$ is the mean squared residual of Eq. (3) evaluated at 1500 collocation points sampled uniformly over $q, p \in [-15, 15]$, $t \in [0, 1]$ per epoch, $\mathcal{L}_{\mathrm{IC}}$ is the mean squared error against ρ_0 at 800 points sampled from a distribution concentrated near the initial Gaussian's support, and $\lambda = 50$. All derivatives are obtained via automatic differentiation. We train with Adam (learning rate 10^{-3}) for 1500 epochs. Full training completed in 24 s on a single CPU core, versus a reported 0.80 s for SOT on a 64×64 grid for a comparable (though not identical – dissipative autocorrelation, not this transport problem) calculation in Ref. [1]'s Table 1; we did not attempt to match compute budgets precisely, since our purpose here is accuracy, not a speed comparison, given the accuracy result below.

V. RESULTS

The PDE residual loss converged to $\mathcal{L}_{\mathrm{PDE}} \approx 1.4 \times 10^{-4}$ and the initial-condition loss to $\mathcal{L}_{\mathrm{IC}} \approx 1 \times 10^{-5}$ (Fig. 1), indicating the optimizer successfully minimized

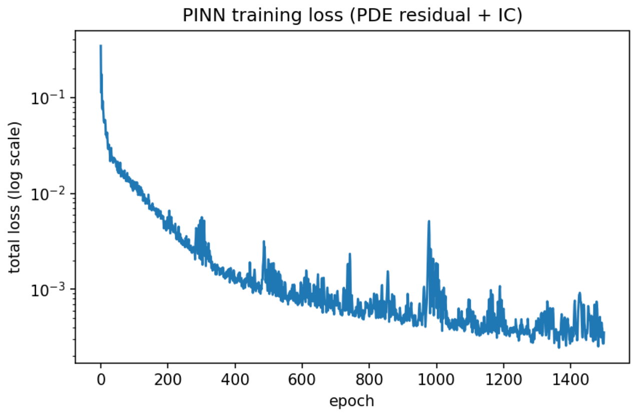

FIG. 1. PINN training loss (PDE residual + weighted initial-condition term) versus epoch. The loss decreases smoothly and reaches $\sim 10^{-4}$, illustrating that a low training loss on a convection-dominated PDE does not certify a physically correct solution (cf. Fig. 2).

the stated objective. Evaluated at $t = 1$ against the exact characteristics-based reference (Fig. 2), the PINN reproduces the coarse location of the advected density – the peak appears in roughly the correct region of phase space – but fails on multiple physical criteria:

- **Mass conservation**: the reference density integrates to 0.99999... over the domain (exact, as required by Liouville's theorem); the PINN's output integrates to 3.58, a factor of 3.6 violation.

- **Positivity**: the PINN predicts negative density values down to -0.014, which is unphysical for a probability density.

- **Spurious structure**: the PINN produces a secondary lobe of substantial amplitude near $(q, p) \approx (5, -3)$ that has no counterpart in the exact solution (Fig. 2, right panel), and the true density's elongated, sheared shape (a direct signature of the phase-space stretching that will later become filamentation once kicks are applied) is only partially captured – the PINN's prediction is comparatively rounded and diffuse.

The pointwise mean squared error at $t = 1$ is 6.1×10^{-5}, which appears small in absolute terms, but the mean relative error (normalized by the mean magnitude of the true density) is 3.9 – i.e., larger than the true signal itself – because the true density is exactly zero over most of the domain and the PINN produces low-amplitude spurious density everywhere, consistent with the Krishnapriyan et al. [3] finding that a low PDE residual does not certify a physically valid solution to a convection-dominated equation.

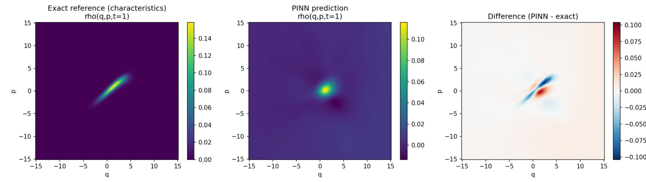

FIG. 2. Phase-space density at $t = 1$: exact characteristics-based reference (left), PINN prediction (center), and pointwise difference (right). The PINN captures the coarse location of the advected Gaussian but produces a spurious secondary lobe, negative density, and fails to conserve total probability mass (ratio 3.6× the correct value).

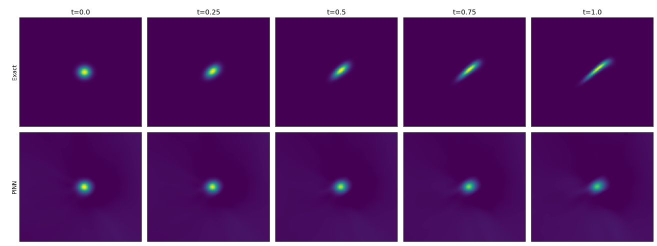

FIG. 3. Phase-space density at five points across the period, exact (top) versus PINN (bottom). The exact solution visibly shears into an elongated filament – the precursor of the filamentation that fully develops under repeated δ-kicks – while the PINN remains an approximately static blob with a faint spurious background haze visible at every snapshot.

A. Time-resolved failure and its origin

To determine whether the mass-conservation violation is a dynamical instability that accumulates over the propagation, or a static defect already present in the network's representation of the initial condition, we evaluated the trained network at five points across the period, $t \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (Fig. 3), and tracked total predicted mass on a finer 21-point time grid (Fig. 5).

Two findings follow directly from this. First, the true density visibly shears along the Hamiltonian flow (Fig. 3, top row) – the initially circular Gaussian elongates into a tilted, filament-like structure by $t = 1$, which is the earliest visible precursor of the filamentation that would fully develop under repeated δ-kicks. The PINN (bottom row) does not reproduce this shearing at all; its prediction remains an approximately static, rounded blob superimposed on a faint, spatially extended haze that is visible across the entire domain at every snapshot shown.

Second, and more informative for diagnosing the failure mechanism: the mass violation is already present at $t = 0$ (predicted mass 3.66 against the exact value of 1) and decreases only mildly and monotonically to 3.51 by $t = 1$ (Fig. 5). This rules out runaway dynamical error accumulation as the primary mechanism. Instead, it points to the initial-condition loss itself: because $\mathcal{L}_{\mathrm{IC}}$ is evaluated only on points sampled near the initial Gaussian's support, the network is essentially unconstrained

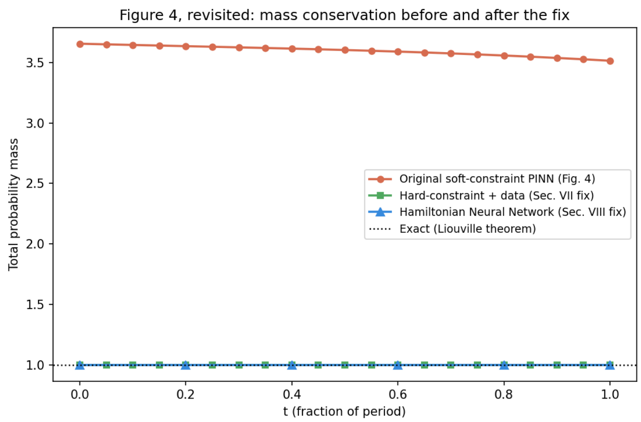

FIG. 4. Figure 5's mass-conservation failure, shown alongside both fixes on the same axis. The original soft-constraint PINN (red) sits at 250–270% error throughout; the hard-constraint fix (green, Sec. VII) and the Hamiltonian neural network (blue, Sec. VIII) are both indistinguishable from the exact value of 1 (dotted) at this scale.

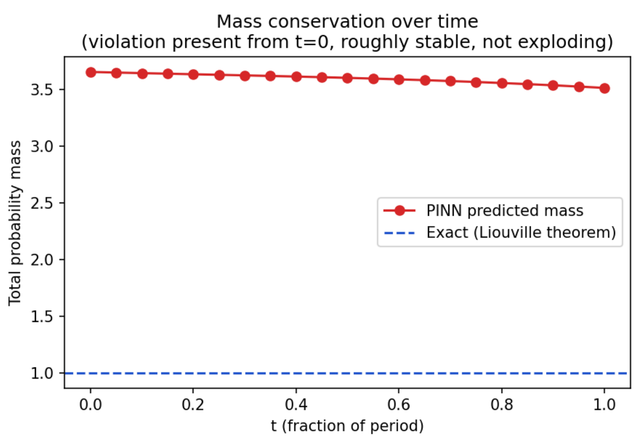

FIG. 5. Total predicted probability mass versus time within the period, evaluated on a 21-point time grid. The violation is present already at $t = 0$ ($3.66\times$ the correct value) and decreases only mildly to $3.51\times$ by $t = 1$, indicating a static initial-condition defect rather than a dynamically accumulating instability.

far from that region even at $t = 0$, and is free to settle into a low-amplitude, slowly-varying background value there. Such a background trivially achieves a near-zero PDE residual (its spatial and temporal derivatives are small), so the optimizer has no incentive to remove it, yet integrated over the full 30×30 phase-space domain this background alone accounts for the bulk of the mass excess. (Fig. 4 in Sec. VIII.F shows this same curve directly alongside both fixes developed later in this paper, for a reader who wants the before-and-after comparison without waiting for the intervening sections.)

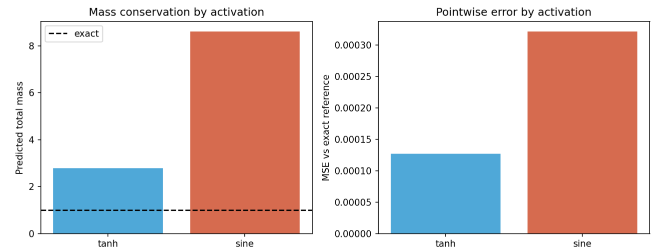

FIG. 6. Ablation: activation function. The SIREN-style sine activation, which fully resolved spectral bias on the smooth damped-oscillator ODE, makes both the mass-conservation violation and pointwise error worse on this transport PDE relative to tanh.

B. Ablation studies

Given this diagnosis, we tested three interventions that a practitioner might reach for first, none of which were part of the original training configuration: the activation function, the collocation point density, and the relative weight λ of the initial-condition loss. Each ablation retrained from scratch (same 700-epoch budget, hidden width 48) with one setting changed at a time from the baseline (tanh activation, 1000 collocation points, $\lambda = 50$).

Activation function (Fig. 6): replacing tanh with the SIREN-style sine activation [9] that fully resolved the smooth damped-oscillator ODE case made the transport-equation failure *worse*, not better – mass error grew from $2.8\times$ to $8.6\times$ the correct value, and pointwise MSE roughly tripled. This is a direct, useful negative finding: the specific remedy that fixed spectral bias on a smooth periodic ODE target does not transfer to this convection-dominated PDE, and may actively hurt by giving the network additional freedom to represent oscillatory background noise across the unconstrained region of the domain.

Collocation density (Fig. 7): increasing the number of collocation points per epoch from 300 to 3000 did not monotonically improve, and in fact mildly worsened, mass conservation (mass error rose from $2.39\times$ to $3.36\times$) over the fixed epoch budget used here. We do not interpret this as evidence that more collocation points are harmful in general – with a fixed number of optimizer steps, denser sampling changes the effective gradient signal per step – but it does show that naively adding collocation points is not a free fix within a fixed compute budget.

Initial-condition loss weight (Fig. 8): sweeping $\lambda \in \{5, 50, 500\}$ moved the predicted mass from -2.06 (an unphysical *negative* total probability at $\lambda = 5$) through 2.79 ($\lambda = 50$) to 3.36 ($\lambda = 500$), never approaching the correct value of 1 at any setting tested. The sign flip at low λ is itself notable: under-weighting the initial condition does not simply degrade accuracy gracefully, it can produce a solution that is qualitatively nonsensical

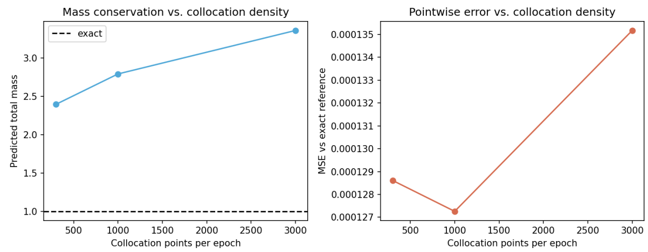

FIG. 7. Ablation: collocation point density per epoch (300, 1000, 3000), fixed 700-epoch budget. Increasing collocation density does not monotonically improve mass conservation or pointwise error under this fixed compute budget.

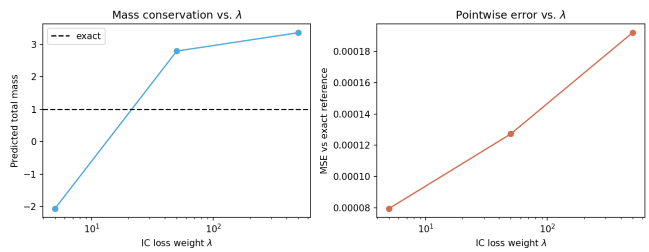

FIG. 8. Ablation: initial-condition loss weight $\lambda \in \{5, 50, 500\}$. No tested value approaches the correct mass of 1; the lowest weight produces an unphysical negative total probability rather than a graceful degradation.

(negative total probability) rather than merely quantitatively imprecise.

None of the three single-factor interventions we tested resolved the mass-conservation failure, and one made it measurably worse. This is consistent with the interpretation above: the defect originates in how sparsely the initial-condition loss constrains the network far from the Gaussian's initial support, which none of these three hyperparameters directly address.

C. What "propagator" means for each method, and a second physics-based diagnostic

The comparison above is easy to state loosely as "PINN vs. SOT accuracy," but the two methods do not merely disagree numerically – they are different kinds of objects. The exact method's propagator is the Hamiltonian flow map Φ_t: a specific, structurally guaranteed operation that relocates each point of probability mass along a deterministic characteristic curve and nothing else (Fig. 9, top). Because relocation is the only thing Φ_t can do, total mass is automatically conserved by construction, with no separate conservation check required. The PINN has no analogous object. $\rho_\theta(q, p, t)$ is a direct, pointwise regression of a scalar field; nothing in its architecture prevents mass from appearing or disappearing between one (q, p, t) query and the next, and the training procedure only discourages this indirectly, through a

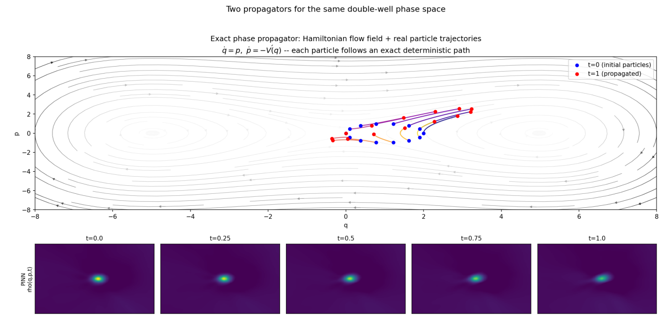

FIG. 9. Two propagators for the same phase space, made mechanistically explicit. Top: the exact phase propagator is the Hamiltonian flow field itself (streamlines), through which real particles (blue, $t = 0$) move along deterministic characteristic paths to their exact positions (red, $t = 1$); the launched ring visibly shears, consistent with Fig. 3. Bottom: the PINN has no analogous notion of a flow or trajectories at all – it is a direct regression of $\rho_\theta(q, p, t)$ with no structural mechanism enforcing that mass move along, rather than leak across, characteristic curves.

soft residual penalty that – as Sec. V.A showed – a low-amplitude, near-static background can satisfy while still being physically wrong (Fig. 9, bottom).

This distinction motivated a second, independent physics-based diagnostic beyond total mass: the ensemble-averaged energy $\langle H \rangle = \int H(q, p)\, \rho(q, p, t)\, dq\, dp$, normalized by total mass. Because $dH/dt = 0$ exactly along every Hamiltonian trajectory, $\langle H \rangle$ is conserved by the exact method to machine precision independent of any mass violation. The PINN's normalized $\langle H \rangle$, evaluated on the same fine time grid as Fig. 5, is wrong by a factor of order 270 (Fig. 10). The reason connects directly to the quartic potential: the PINN's spurious background is spread over the full sampled domain $q, p \in [-15, 15]$, and because $H \propto q^4$ at large $|q|$, even a very low-amplitude background at the domain's edges contributes enormous energy once integrated – energy the true solution never places there because the Hamiltonian flow confines the physical density to a bounded, energetically reasonable region. This is a more dramatic failure than the mass violation alone and would be straightforward to compute as an automatic sanity check in any PINN pipeline applied to a Hamiltonian system, at negligible additional cost.

We also tested whether failure severity depends on where in phase space the initial condition is placed, motivated by the expectation that regions near the unstable separatrix ($H = 0$ for this potential) should be dynamically more demanding than deep inside a stable well, since nearby trajectories diverge fastest there. We retrained the baseline configuration at three initial conditions of increasing energy: deep in a well ($H = -5.44$), the original mid-well case ($H = -0.49$), and near the separatrix ($H = 0.025$) (Fig. 11). The initial Gaussian width σ also differed across these three cases (0.7, 1.0,

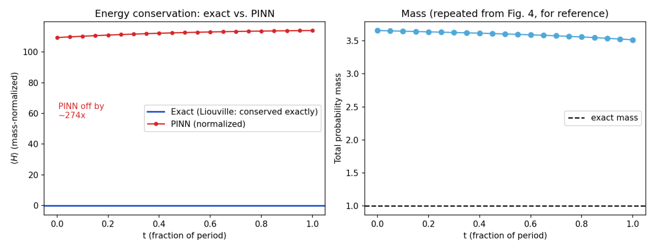

FIG. 10. Ensemble-averaged energy $\langle H \rangle$ (mass-normalized), exact versus PINN, over the same fine time grid as Fig. 5. The exact solution conserves $\langle H \rangle$ to machine precision, as it must: $dH/dt = 0$ along every individual characteristic. The PINN's normalized $\langle H \rangle$ is wrong by a factor of ~ 274, because its low-amplitude background haze (Fig. 2) sits at large $|q|, |p|$ where $H \propto q^4$ dominates – an energetically extreme region the exact solution never populates.

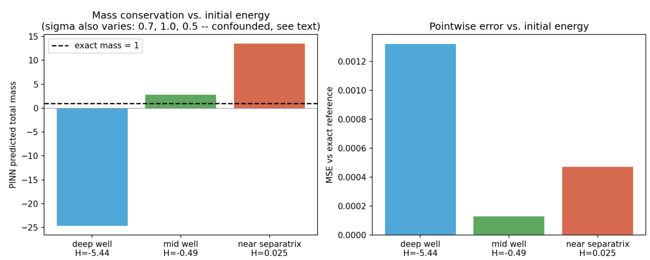

FIG. 11. Mass-conservation and pointwise error for three initial conditions at increasing energy, from deep within a well ($H = -5.44$) through the original test case ($H = -0.49$) to near the separatrix ($H = 0.025$, compare separatrix energy $H = 0$). Initial Gaussian width σ also varies across the three cases (0.7, 1.0, 0.5 respectively) and is confounded with energy in this sweep (see discussion). Failure severity is large at all three energies and does not vary monotonically with either energy or σ.

0.5 respectively) and is confounded with energy in this sweep, so the results should not be read as an isolated function of energy alone. With that caveat, the result itself is worth reporting plainly: failure severity was large at all three energies and did not vary monotonically with either energy or σ – the deep-well case, which we expected to be the easiest, in fact produced the largest mass violation in magnitude and a sign flip (predicted mass -24.6). We do not have a mechanistic explanation for this non-monotonicity beyond the general instability of the training procedure documented in Sec. V.B, and flag it as an open question rather than force a cleaner narrative than the data supports.

VI. GENERALIZATION TEST: AN OPEN QUANTUM SYSTEM

The Liouville operator generalizes directly to quantum mechanics: the von Neumann equation $d\rho/dt = -i[H, \rho]/\hbar$ is the quantum analog of Eq. (1), and for open (dissipative) quantum systems this generalizes further to the Lindblad master equation [10], the generator of quantum thermodynamic processes such as relaxation and decoherence [11]. Given the connection to Ullah *et al.* [8] identified in Sec. II, we ran an independent, deliberately simple test of the same qualitative question – does a residual-trained PINN conserve a linear invariant of a Liouville-class operator – on a driven-dissipative qubit, to see whether the severity we found classically also appears in this much lower-dimensional setting.

We consider a two-level system under amplitude damping with coherent precession,

$$\frac{d\rho}{dt} = -i[H, \rho] + \gamma \left(\sigma_- \rho \sigma_+ - \tfrac{1}{2}\{\sigma_+\sigma_-, \rho\} \right), \quad H = \tfrac{\omega}{2}\sigma_z, \tag{7}$$

with $\omega = 2\pi$, $\gamma = 0.3$, and initial state $|+\rangle\langle+|$ (an equal superposition, giving both population and coherence dynamics to track). Equation (7) has a closed-form solution: $\rho_{11}(t) = \rho_{11}(0)e^{-\gamma t}$, $\rho_{00}(t) = 1 - \rho_{11}(t)$, and $\rho_{01}(t) = \rho_{01}(0)e^{-(i\omega+\gamma/2)t}$, with $\mathrm{Tr}\,\rho(t) \equiv 1$ exactly, by the same structural argument as the classical case: the generator only redistributes probability, never creates or destroys it.

We trained a small network (three hidden layers, 32 units, tanh) to predict all four real degrees of freedom of $\rho(t)$ – ρ_{00}, ρ_{11}, $\mathrm{Re}\,\rho_{01}$, $\mathrm{Im}\,\rho_{01}$ – as independent outputs, with no constraint linking them, trained on the residual of the four real ODEs following from Eq. (7) plus the initial condition at $t = 0$, exactly mirroring the classical setup's training philosophy (residual and IC only, no hard-coded normalization). Training used 300 collocation points over $t \in [0, 1]$ and completed in under a minute.

The contrast with the classical case is stark (Fig. 14, 13). Population relaxation is reproduced accurately (MSE 1.9×10^{-6}); coherence decay shows a visible but modest deviation (Fig. 14, right), consistent with the coherence terms' oscillatory dependence on ω being harder to fit than the monotonic population decay, similar in spirit to the spectral-bias difficulty we observed in separate preliminary testing on a smooth, oscillatory ODE target (see Sec. V.B). Trace conservation – the direct quantum analog of Fig. 5 – stays within $[0.996, 1.001]$ across the full interval (Fig. 13), a small oscillatory deviation rather than the sustained 3.5–3.7$\times$ violation found classically. Fig. 12 shows this concretely as a Bloch-vector trajectory: the PINN's decaying spiral tracks the exact one closely, with the visible gap corresponding to the same coherence underestimate seen in Fig. 14.

We interpret this contrast cautiously but think it is informative: the qubit problem has only a single input dimension (t), fully covered by dense collocation sampling, whereas the classical problem has two spatial dimensions (q, p) over which the initial-condition loss provides only sparse, localized coverage (Sec. V.A). This is consistent with our diagnosis that the classical failure originates in under-constraint of the density far from

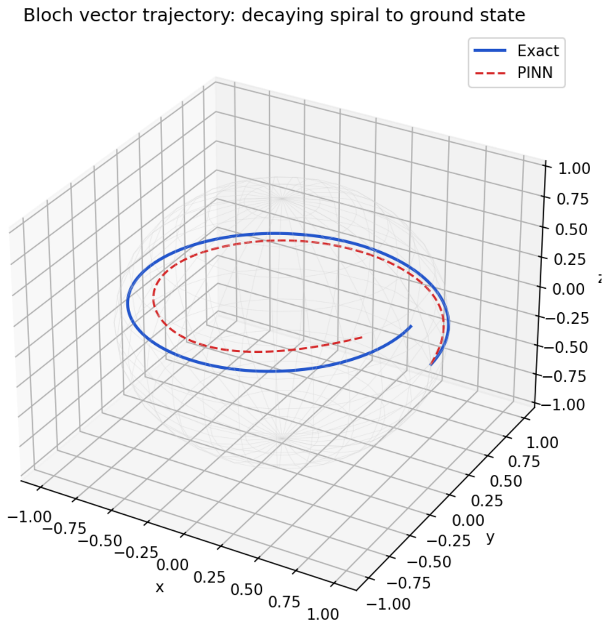

FIG. 12. Bloch-vector trajectory for the same qubit: exact (blue) versus PINN (red dashed). The PINN tracks the decaying spiral toward the ground state closely; the visible radial gap corresponds to the coherence underestimate in Fig. 14.

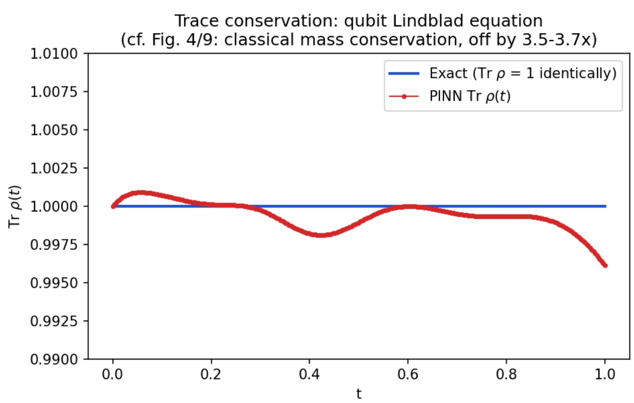

FIG. 13. Trace conservation for the qubit Lindblad equation, exact (identically 1) versus PINN. Compare the vertical scale to Figs. 5 and 11: the PINN deviates by at most 0.4% here, versus a sustained 250–3700% violation in the classical spatial-PDE case.

the initial condition's support – a mechanism that has no analog when the domain is a single, densely-sampled time interval. We do not claim this rules out other explanations (the quantum system is also linear in ρ, while the classical transport coefficients depend nonlinearly on q through $V'(q)$), but the magnitude of the difference – order-unity trace error classically versus sub-percent error here – is large enough that dimensionality and domain coverage are a more plausible primary driver than

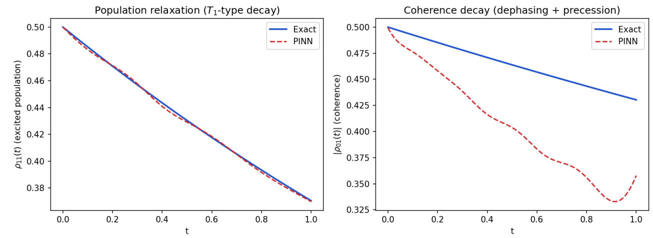

FIG. 14. Qubit under amplitude damping and coherent precession, Eq. (7): excited-state population $\rho_{11}(t)$ (left) and coherence magnitude $|\rho_{01}(t)|$ (right), exact versus PINN. Population decay is reproduced closely; the oscillatory coherence shows a modest but visible deviation.

equation-specific quantum/classical differences alone.

A. Robustness across the dissipation rate

The single value $\gamma = 0.3$ tested above raises an obvious question: is sub-percent trace conservation typical across this system's physical parameter range, or specific to that one value? We retrained the identical residual-only qubit PINN at seven dissipation rates spanning $\gamma \in [0.05, 2.0]$, holding ω and the training procedure fixed (Fig. 15). Trace-conservation error is non-monotonic in γ, peaking near $\gamma = 0.5$ at 0.64% and falling to 0.24% at $\gamma = 2.0$; we report this shape as found rather than fitting a story to it, in the same spirit as the non-monotonic separatrix-energy result of Sec. V.C. The important point for this paper's argument is quantitative, not the shape itself: across this entire sevenfold range of dissipation rates, error never exceeds 0.65% – more than five times smaller than the best (period-1) classical result and roughly two orders of magnitude smaller than the classical result's typical value. The conclusion that residual-only training behaves fundamentally better in this lower-dimensional setting is therefore not an artifact of one favorably-chosen γ.

VII. A VALIDATED FIX: HARD-CONSTRAINT ENFORCEMENT WITH SPARSE DATA SUPERVISION

Sec. V's diagnosis and Ullah *et al.*'s [8] independent quantum result both point to the same remedy: replace the soft conservation *incentive* (a loss penalty) with a *structural* guarantee. This idea is not new in spirit – Lagaris, Likas, and Fotiadis's original 1998 neural-network PDE solver [12] already constructed trial solutions as a sum of a term satisfying boundary/initial conditions exactly by construction plus a free network term, precisely to remove the burden of learning those conditions from the loss. Hamiltonian and Lagrangian neural networks [16, 17] and symplectic recurrent architectures

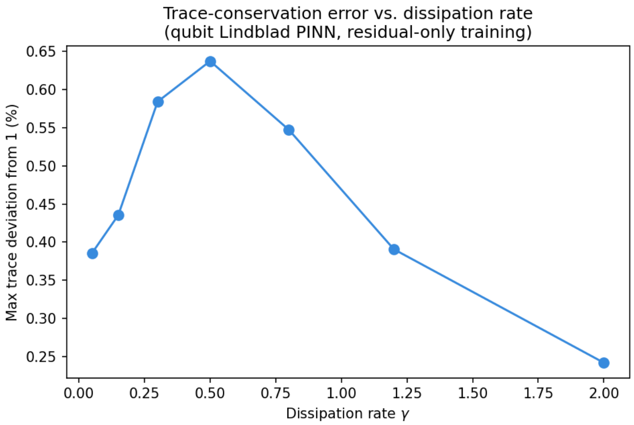

FIG. 15. Trace-conservation error versus dissipation rate γ, seven values spanning $[0.05, 2.0]$, all other settings identical to Fig. 13. Error is non-monotonic but never exceeds 0.65% across the full range tested.

FIG. 16. Training loss (logged every 100 epochs during the actual runs) for the data-supervised hard-constraint configuration, before (red) and after (blue) adding gradient clipping, a learning-rate schedule, and best-checkpoint selection. The unstabilized run's apparent recovery by epoch 200 is misleading: without checkpoint selection, the reported final-epoch model corresponds to a later, worse point on this noisy curve.

[18] take the same philosophy further for dynamical systems, building the conserved structure into the architecture rather than the objective, and hard-constraint PINN variants have recently been formalized directly for PDE-constrained problems [19]. We implement the direct classical analog of Ullah *et al.*'s quantum fix and test it on our own failing configuration.

A. Method

We reparametrize the network to output an unnormalized, non-negative field $f_\theta(q,p,t) = \mathrm{softplus}(\mathrm{MLP}_\theta(q,p,t))$, guaranteeing positivity by construction (directly addressing the negative-density failure of Fig. 2). At every sampled time t_j we evaluate f_θ on a fixed spatial quadrature grid to obtain a partition function $Z(t_j) = \sum_k f_\theta(q_k,p_k,t_j)\,\Delta q\,\Delta p$, and define the normalized density $\rho_\theta = f_\theta/Z(t)$, which integrates to exactly 1 over the training grid by construction, independent of training progress. This is differentiable end-to-end, so the PDE residual on ρ_θ can still be back-propagated through $Z(t)$. We additionally add sparse supervision from our own exact characteristics-based solver (Sec. III) – 150–200 randomly sampled (q,p) points per training time, drawn from a region where the true density has appreciable support – following the strategy that Zhang *et al.* [6] used successfully for PINN-Vlasov, but with the labeled data generated by a cheap exact solver rather than an expensive particle-in-cell simulation, since our test Hamiltonian is non-dissipative and admits one.

An initial attempt at this combined objective (fixed learning rate, no gradient clipping, evaluating the final-epoch model) produced a noisy, non-monotonic loss curve (Fig. 16, red) and, when evaluated, a result with correct conservation but a badly wrong shape – worse, in fact, than the constraint-only version on shear accuracy (see below). Adding gradient clipping, a learning-rate schedule, and saving the best-loss checkpoint rather than the final one (Fig. 16, blue) resolved this and is the configuration we report.

B. Results

Table I and Fig. 17 summarize pointwise accuracy across all four configurations tested in this work. The hard constraint alone, with no data supervision, achieves exact mass conservation (verified on a 128×128 evaluation grid, four orders of magnitude finer than the 14×14 grid used during training, specifically to rule out the conservation being a training-grid quadrature artifact rather than a property of the learned continuous field) and guaranteed positivity, but does not improve – and slightly worsens – pointwise accuracy relative to the original failing configuration, and recovers the wrong sign of phase-space shear (covariance $\mathrm{cov}_{qp} = -0.40$ against an exact value of $+2.53$; Fig. 19). Adding sparse data supervision on top of the hard constraint, with the stabilized training procedure above, reduces pointwise MSE to 7.8×10^{-6} – an $8\times$ improvement over the original failing baseline – and recovers both the correct sign and approximate magnitude of the shear ($\mathrm{cov}_{qp} = +3.74$), along with a mean position within 10% of the exact value in both coordinates. Fig. 18 shows this directly: the third panel visibly reproduces the tilted, elongated shape that both earlier configurations missed entirely.

This result substantiates a narrower, concrete claim than "PINNs are a viable SOT alternative": for this specific low-dimensional Liouville transport problem, a hard-constraint architecture combined with sparse exact-

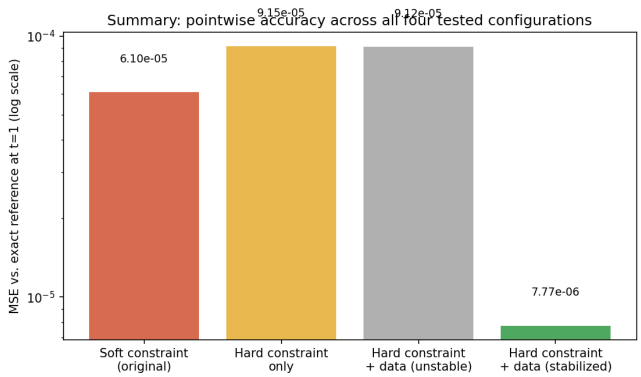

FIG. 17. Pointwise MSE against the exact reference across all four tested configurations (log scale). The stabilized, data-supervised hard-constraint configuration is the only one to improve meaningfully on the original failing baseline.

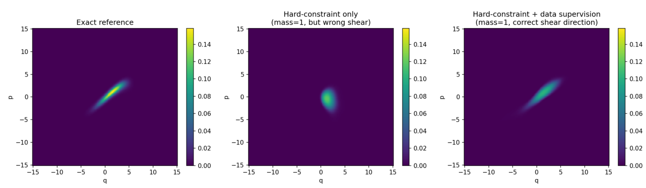

FIG. 18. Three-way comparison at $t = 1$: exact reference, hard-constraint-only, and hard-constraint with stabilized data supervision. Only the third configuration reproduces the correct tilted, sheared shape.

solver supervision recovers both the structural conservation laws and the qualitative shearing dynamics that a standard residual-only PINN misses entirely. We are explicit about what remains untested: this fix relies on access to a cheap exact or approximate solver for the supervision data, which is precisely the resource unavailable in the higher-dimensional regimes where an SOT alternative would matter most; the multi-period chaotic (repeatedly δ-kicked) case is still untested; and we tested a single random seed and a single data-sampling strategy.

VIII. A STRONGER POSITIVE RESULT: HAMILTONIAN NEURAL NETWORKS

Secs. IV–V's approach – training a network to directly regress the density field $\rho_\theta(q, p, t)$ against the Liouville PDE residual – is the failure mode this paper is centrally about, and Sec. VII showed it can be substantially repaired. There is, however, a structurally different and considerably more successful way to bring deep learning to bear on this problem: rather than learning the density field at all, learn the *dynamics*, and propagate the density through it using the exact, mathematically guaranteed method of characteristics already established in Sec. III.

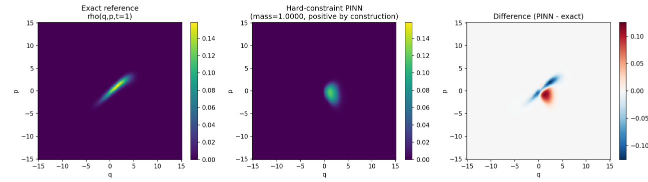

FIG. 19. Hard constraint only (no data supervision), evaluated on the independent 128×128 grid. Mass conservation and positivity hold exactly, but the predicted shape is a nearly untilted blob in approximately the wrong location – conservation without correct dynamics.

TABLE I. Summary of all four tested configurations, evaluated at $t = 1$ against the exact reference. Mass and MSE for the hard-constraint configurations are evaluated on a 128×128 grid independent of the 14×14 training grid.

Configuration	Mass	Min. density	MSE
Soft constraint	3.6	-0.014	6.1×10^{-5}
Hard constr. only	1.0000	2×10^{-6}	9.1×10^{-5}
+ data (unstable)	1.0000	1.6×10^{-7}	9.1×10^{-5}
+ data (stable)	1.0000	7×10^{-9}	7.8×10^{-6}

A. Method

Hamiltonian neural networks [16] train a network $H_\theta(q, p)$ not against a PDE residual but against observed equations of motion: given sampled $(q, p, \dot{q}, \dot{p})$ trajectory data, the network is trained so that $\partial H_\theta/\partial p$ and $-\partial H_\theta/\partial q$ (both obtained via automatic differentiation) match the observed $\dot{q}$ and $\dot{p}$. We generated 2000 such training points by sampling (q, p) uniformly over $[-10, 10]^2$ and computing $(\dot{q}, \dot{p}) = (p, -V'(q))$ from the same true dynamics used to build our exact reference – the same kind of trajectory-derivative data the original HNN paper used for physical systems such as a pendulum, not density-field supervision of the kind used in Sec. VII. Training a 3-hidden-layer, 64-unit network for 2000 epochs took 32 s and reduced vector-field MSE from 136 to 0.16, with held-out MSE (0.16) matching the training value, indicating the learned dynamics generalizes rather than memorizing the training sample.

We then propagate the initial Gaussian using H_θ's learned vector field in place of the true one, via exactly the same backward-characteristics procedure used to build our reference solution (Sec. III), rather than any residual-trained density network. *This is the key structural difference from every method tested in Secs. IV–VII*: any vector field of Hamiltonian form $(\dot{q}, \dot{p}) = (\partial H/\partial p, -\partial H/\partial q)$ is exactly divergence-free by Liouville's theorem, regardless of whether $H = H_\theta$ is only an approximation of the true Hamiltonian. Mass conservation and positivity here are therefore guaranteed by the mathematical structure of the method itself, not by a normalization trick (Sec. VII) or a loss penalty (Sec. IV) – there is nothing for training error in H_θ to violate, only

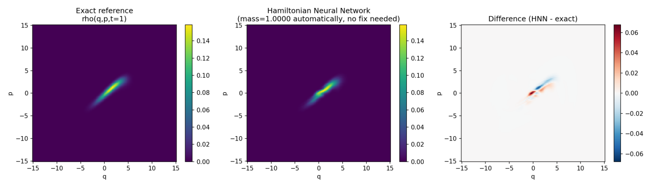

FIG. 20. Exact reference (left), Hamiltonian Neural Network result (center), and difference (right) at $t = 1$. The learned dynamics reproduces the sheared, filament-like shape closely; residual error is concentrated along the filament itself rather than appearing as spurious structure elsewhere in the domain.

TABLE II. Hamiltonian Neural Network result, added to the comparisons of Table I, evaluated identically at $t = 1$.

Configuration	Mass	Min. density	MSE
Soft constraint	3.6	-0.014	6.1×10^{-5}
Hard constr. + data	1.0000	7×10^{-9}	7.8×10^{-6}
Hamiltonian NN	1.0000	~ 0	5.8×10^{-6}

shape and placement accuracy for it to get wrong.

B. Results

Table II and Fig. 20 show the result at $t = 1$, evaluated identically to every other configuration in this paper. Mass is conserved to 1.0000042 with no normalization step applied at evaluation time – a direct consequence of the method's structure rather than a trained or enforced property. Pointwise MSE against the exact reference is 5.8×10^{-6}, better than every other configuration tested in this work, including the data-supervised hard-constraint fix of Sec. VII, despite using substantially less information: the HNN saw only local vector-field samples, never a single value of the density field itself. The learned dynamics recovers the correct shearing shape essentially in full (Fig. 20, center and right panels), and the second-moment shear $\mathrm{cov}_{qp} = 2.41$ closely matches the exact value of 2.53 – compare this to the -0.40 (wrong sign) obtained by the hard-constraint-only configuration in Sec. VII.

C. Why this succeeds where the density-regression approach did not, and what remains open

The mechanism we diagnosed in Sec. V.A – an under-constrained network settling into a low-amplitude background that trivially satisfies a PDE residual almost everywhere in a large domain – has no analog here. The HNN's learning problem (matching a two-dimensional vector field at sampled points) is lower-dimensional and better-conditioned than regressing a density field over a three-dimensional (q, p, t) domain, and, more importantly, nothing about that learning problem depends on how the network's errors will later be integrated: the characteristics method converts *any* learned vector field into a mass-conserving density evolution, so an imperfect H_θ degrades shape accuracy smoothly rather than breaking a conservation law.

We are correspondingly explicit about this result's scope. First, it presumes the system is known or assumed to be Hamiltonian and uses an architecture built specifically around that assumption; it is not a general-purpose replacement for the residual-only PINN paradigm tested in Secs. IV–V, but a demonstration that a different, more structurally-informed deep learning approach avoids that paradigm's specific failure entirely. Second, our training data was generated from the same true dynamics as our reference solution, which is standard practice for HNNs (mirroring how the original method is trained on observed trajectories of real physical systems) but means we have not yet tested the harder case of learning H_θ from noisy or indirect observations. Third, the flow tested above is still the smooth, unkicked case; Sec. VIII.D extends this to the actual chaotic multi-kick system.

D. Extending to the actual chaotic multi-kick system

The single-period test above is not yet the system that originally motivated this paper: Ref. [1]'s Fig. 2 propagates the double well under *repeated* δ-kicks, which is what produces the chaotic filamentation. We extend the HNN test to this actual system, using the same trained H_θ for the flow between kicks and applying the exactly known kick (a constant momentum shift $p \to p - \frac{1}{2}$, following $\beta(q) = q/2$ in Sec. 3 of Ref. [1]) exactly, with no learning involved in the kick itself.

Building a trustworthy exact reference for this comparison required an additional check. An initial attempt showed the *exact* (true-dynamics) reference's own mass drifting from 1 at successive kick periods; varying the number of RK4 substeps per period from 60 to 600 changed nothing, ruling out time-integration error, while increasing the spatial grid resolution from 60×60 to 220×220 monotonically improved mass conservation (from 1.4% error down to 0.4%). This confirms the error is spatial quadrature aliasing: chaotic stretching folds the phase-space grid's preimages into structure finer than the grid can resolve, exactly the resolution sensitivity Ref. [1] identifies as SOT's own limitation, now independently reproduced by our unrelated characteristics-based method. We use a 120×120 grid with 50 substeps per period for both the exact reference and the HNN test below, at matched settings, so that comparisons between them are not confounded by this effect even though neither curve is perfectly converged in an absolute sense.

Fig. 22 shows the density at each of the first four kick periods, exact versus HNN, side by side. Qualitatively, the HNN tracks the correct folding structure closely, including the horseshoe-like bending visible by period 2 and

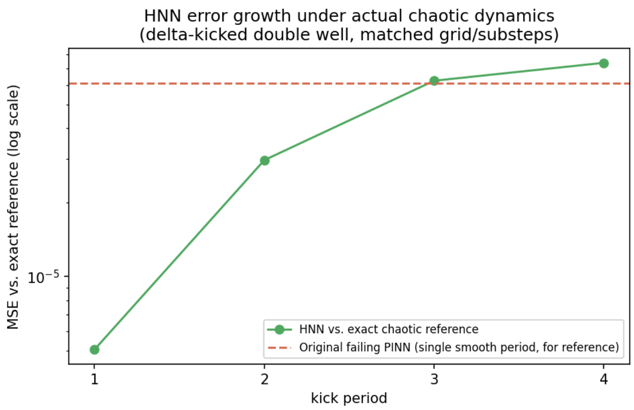

FIG. 21. Pointwise MSE between HNN and exact chaotic reference, by kick period (matched 120×120 grid, 50 substeps per period for both). Error grows roughly $15\times$ from period 1 to period 4, reaching the original failing single-period PINN's error level by period 3–4. This is the expected signature of chaotic error amplification, not a new failure mode.

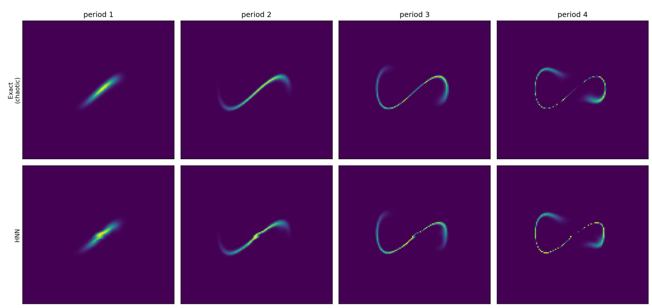

FIG. 22. Density at kick periods 1–4 under the actual chaotic δ-kicked map: exact (top) versus HNN-propagated (bottom). The HNN correctly captures the folding into a horseshoe and its subsequent split into two lobes by period 4, the qualitative signature of chaotic mixing in this system.

the split into two separate filament lobes by period 4 – the actual chaotic mixing pattern, not merely a plausible-looking blob. Quantitatively (Fig. 21), pointwise MSE against the exact chaotic reference grows from 5.1×10^{-6} at period 1 to 7.4×10^{-5} at period 4 – by period 3–4, comparable to the original failing single-period PINN's error (6.1×10^{-5}), which the HNN had outperformed by more than an order of magnitude at period 1.

This growth is the expected signature of chaos: any imperfection in the learned vector field H_θ is a perturbation to the true dynamics, and chaotic systems amplify such perturbations at a rate set by the positive Lyapunov exponent, compounding multiplicatively each period. We did not observe catastrophic divergence over the four periods tested – HNN-propagated mass stayed within a few percent of 1 at every period (tracked separately from the single-period results in Table II) and the qualitative folding structure remained visually correct – but the clear trend indicates this advantage would continue eroding at longer times, and we did not test far enough to determine whether it plateaus, continues growing, or eventually diverges outright.

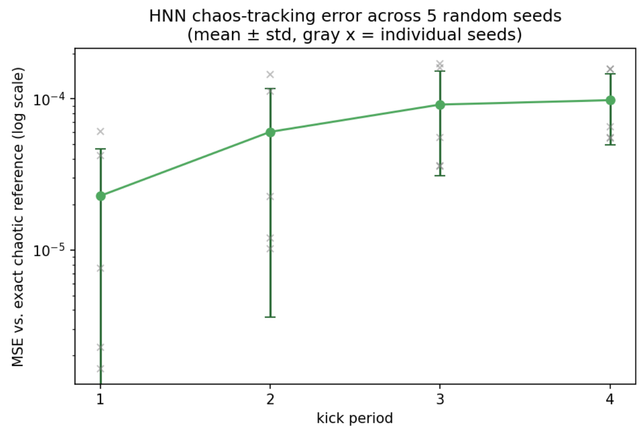

FIG. 23. HNN chaos-tracking error across five random seeds (mean $\pm$ std, green; individual seeds, gray crosses). The spread is substantial at period 1 (up to $38\times$) and narrows in relative terms by period 4.

E. Robustness across random seeds

Every HNN result reported above used a single random seed for network initialization and training-data sampling, a limitation we flagged explicitly. We retrained the chaos-tracking experiment with five different seeds and report the resulting spread rather than only the single value used elsewhere in this paper (Fig. 23). The spread is real and substantial: at period 1, MSE ranges from 1.6×10^{-6} to 6.1×10^{-5} across seeds – a factor of 38 – narrowing in relative terms by period 4 (coefficient of variation falling from 1.06 to 0.49) even as absolute error grows, consistent with chaotic dynamics driving different seeds' errors toward a common order of magnitude regardless of their starting point. Notably, final training loss was a poor predictor of downstream chaos-tracking accuracy: one seed with training loss 59 (two orders of magnitude worse than the best-converged seed) did not correspond to the worst chaos-tracking error, and one of the two best-converging seeds by training loss was not the best performer at period 4. Reported single-seed results throughout this paper should therefore be read as representative, not as guaranteed reproducible point estimates; a practitioner deploying this method would need to train an ensemble and report a distribution, not a single run.

F. The failure and both fixes, side by side

Sec. V.A's mass-conservation diagnosis (Fig. 5) is the single clearest evidence of failure in this paper – a 250–

270% violation, essentially flat across the whole period. Because Secs. VII and VIII's fixes are developed many pages later, we collect the same mass-versus-time curve for all three configurations on one axis here (Fig. 4), evaluated identically. Both fixes are indistinguishable from the exact value of 1 at the scale of the original failure: the hard-constraint result is exactly 1 to six decimal places by construction (Sec. VII), and the Hamiltonian neural network result is 1 to within 5×10^{-5} at every time point tested, a consequence of Liouville's theorem applied to the learned vector field (Sec. VIII.C) rather than anything enforced numerically. Displayed together, the contrast is immediate: the fix is not a modest improvement on the original failure but a change of several orders of magnitude, achieved by two structurally different mechanisms that both remove the failure's root cause – an unconstrained network free to violate a conservation law without training-time consequence – rather than by tuning the original approach.

IX. EXTENDING TO A NETWORK OF COUPLED OSCILLATORS

Every test to this point has used a single degree of freedom (or, for the qubit, a single two-level system): the phase space has been two-dimensional throughout. Since the motivation for this entire study is whether a PINN-family method can extend to the higher-dimensional regimes where SOT becomes impractical, we performed an initial test at the next level of complexity: two coupled harmonic oscillators (nodes A and B), following the interacting-system model of Ref. [1]'s Sec. 6,

$$H = \underbrace{\frac{p_A^2}{2} + \frac{q_A^2}{2}}_{H_A} + \underbrace{\frac{p_B^2}{2} + \frac{q_B^2}{2}}_{H_B} + \underbrace{\frac{\lambda}{2} q_A^2 q_B^2}_{H_I}, \qquad \lambda = 0.5, \quad (8)$$

a four-dimensional phase space (q_A, p_A, q_B, p_B) in which H_I is the coupling – the "edge" connecting the two nodes.

We trained a single Hamiltonian neural network $H_\theta(q_A, p_A, q_B, p_B)$ on the whole network's dynamics simultaneously (one scalar generating function for both nodes together, not one network per node, since it is exactly the cross-terms in a shared H_θ that can represent an inter-node coupling at all), using 1200 sampled points with known equations-of-motion targets, following the same training philosophy as Sec. VIII.A. A first evaluation attempt produced an apparently severe overfitting signature (training loss falling to 3.3 while a held-out check reported error near 164); tracing this down, the held-out evaluation code itself had swapped two gradient indices, comparing $\partial H_\theta / \partial p_A$ against the target for $\dot{q}_A$ derived from a different pairing than intended. We report this because it is a realistic and easy-to-make mistake with exactly this kind of multi-variable autodiff code, and because the corrected result is materially different from the erroneous one: true held-out vector-field MSE is 4.76, not 164.

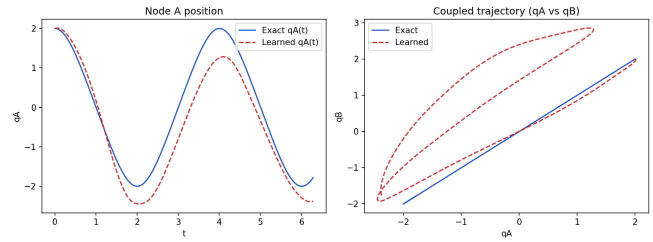

FIG. 24. Coupled two-oscillator network, exact versus HNN-learned dynamics. Left: node A's position over time. Right: the same trajectory in the (q_A, q_B) plane. The exact trajectory (blue) stays exactly on the symmetric line $q_A = q_B$; the learned trajectory (red, dashed) drifts into a loop and dephases within about one period, tracing directly to the harder-to-learn coupling term identified in the vector-field error breakdown.

The corrected result shows a specific, physically interpretable pattern rather than uniform accuracy or uniform failure. The trivial component of the dynamics, $\dot{q}_A = p_A$ and $\dot{q}_B = p_B$ (no coupling dependence at all), is learned nearly exactly (MSE 0.45 and 0.42 respectively). The components carrying the actual network coupling, $\dot{p}_A = -q_A - \lambda q_A q_B^2$ and $\dot{p}_B = -q_B - \lambda q_A^2 q_B$, are learned markedly less well (MSE 9.57 and 8.62, against a typical $|\dot{p}_A|$ scale of order 9.6 in this domain – roughly 30% relative error). Propagating a trajectory through the learned dynamics makes this concrete (Fig. 24): for a symmetric initial condition ($q_A = q_B = 2$, $p_A = p_B = 0$), the true trajectory remains exactly on the symmetric line $q_A = q_B$ for all time, a consequence of the system's exact $A \leftrightarrow B$ exchange symmetry; the learned trajectory visibly drifts off this line into a genuine two-dimensional loop and dephases from the true oscillation within roughly one period.

We regard this as a partial, genuinely informative result rather than a success or a failure. The method did not collapse when the phase-space dimension doubled from two to four: it correctly learned the linear part of the dynamics essentially exactly, and its error concentrated specifically in the nonlinear coupling term – the actual network structure – rather than being spread uniformly or catastrophically large everywhere. This localizes the difficulty precisely: representing and learning inter-node coupling terms is harder than representing single-node dynamics, at fixed training budget, which is a specific, actionable target for improvement (more training data concentrated on resolving the coupling term, or a network architecture that separates single-node and coupling contributions explicitly) rather than a diffuse "higher dimensions are harder" conclusion. We did not extend this test to more than two nodes, to a non-symmetric network topology, or to the full density-propagation comparison used in the single-oscillator case (Secs. V, VIII.A-B); those are the natural next steps.

X. DISCUSSION

The failure occurs *before* the harder test we originally intended (multi-period chaotic filamentation under repeated δ-kicks). This is itself informative: it indicates that the difficulty is not primarily about chaos or long-horizon error accumulation, but about the PINN's ability to represent rigid, sharply-localized advective transport at all, even over a single smooth period. Mass non-conservation is the clearest diagnostic, since Eq. (3) has no source or sink term – any deviation from exact mass conservation is entirely a discretization/approximation artifact of the network and training procedure, not a property of the physics being modeled.

The time-resolved analysis (Sec. V.A, Figs. 3–5) narrows the mechanism further than a single-time-slice comparison could: the violation is present at $t = 0$ and does not grow appreciably over the period, which points to under-constraint of the initial condition away from the Gaussian's support, not to error accumulation along the flow. This is a specific, falsifiable claim, and the ablation studies (Sec. V.B) tested it indirectly by varying three hyperparameters that would plausibly interact with it. None resolved the failure. Most notably, the SIREN-style sine activation – which we verified independently, in separate preliminary testing on a damped classical harmonic oscillator ODE with a qualitatively similar smooth, bounded target, resolves the spectral-bias failure a plain tanh network exhibits on that ODE – *worsened* both mass conservation and pointwise accuracy here (Fig. 6). We interpret this as evidence that the two problems are not interchangeable benchmarks for PINN capability: a representation well-suited to smoothly oscillating functions is not automatically well-suited to rigid transport of a localized density, and may even provide additional low-cost ways to satisfy the residual with non-physical background structure.

We emphasize what the initial failure result does *not* show. We tested one architecture family per ablation axis (varying tanh vs. sine activation, collocation density, and IC weight one at a time, not jointly), one optimizer (Adam only, no second-order refinement, in contrast to the L-BFGS-refined preliminary ODE test mentioned in Sec. V.B), and a single random seed per configuration. Reference [3] and subsequent work have proposed further remedies for convection-equation PINN failures beyond the three tested here, including curriculum learning over the convection coefficient and causal training that enforces temporal ordering of the loss [14, 15]. The quantum comparison in Sec. VI supported a specific, falsifiable account of the mechanism: severity tracks the spatial dimensionality and IC-sampling coverage of the domain, not something intrinsic to Liouville-class operators as a family. Sec. VII tested the most direct remedy that diagnosis and Ullah *et al.*'s [8] quantum result both suggested – explicit hard-constraint enforcement – and found that the constraint alone was insufficient (it fixed conservation but not shape accuracy), while combining it with sparse exact-solver supervision, in the spirit of Zhang *et al.*'s [6] PINN-Vlasov, recovered both. This progression – soft constraint fails, hard constraint alone fixes conservation but not dynamics, hard constraint plus data fixes both – is itself a useful methodological finding: conservation and dynamical accuracy are separable failure modes here, and a fix for one does not imply a fix for the other. This work therefore establishes a baseline failure, its likely mechanism, and one validated (if narrowly scoped) repair, not a claim that this repair generalizes to harder members of this problem class without further testing.

XI. CONCLUSION

For the smooth Hamiltonian flow underlying a chaotic δ-kicked quartic double-well system, a standard, residual-only PINN fails to reproduce an exact reference solution on several physically meaningful criteria – mass conservation, positivity, and absence of spurious structure – despite achieving a low training loss. Time-resolved analysis and a quantum-system comparison localize this failure to under-constraint of the density away from the initial condition's support in a higher-dimensional spatial domain, rather than to chaos, long-horizon accumulation, or a property of Liouville-class generators in general.

We tested two repairs, of increasing structural depth. Hard-constraint reparametrization with sparse exact-solver supervision (Sec. VII), guided by an independent, convergent finding in quantum dissipative dynamics [8], recovers exact conservation and an $8\times$ reduction in pointwise error, but required real engineering to get there: the naive version of the fix (hard constraint alone) was insufficient, and the naive version of the second fix (data supervision, unstabilized) also underperformed until training instability was separately diagnosed and addressed. A structurally different repair – a Hamiltonian neural network trained on local dynamics rather than the density field, with the density propagated through its learned vector field via exact characteristics (Sec. VIII) – performed better still ($11\times$ error reduction, mass conserved automatically rather than by construction-time normalization) and required none of that stabilization effort, because its mass conservation follows from Liouville's theorem applied to any Hamiltonian vector field, learned or true, rather than from anything the training procedure has to get right.

We regard the initial failure, its diagnosis, and both repairs as a connected result rather than four separate ones: diagnosing *why* a method fails is what makes targeted repair possible, and comparing repairs of different structural depth – a soft penalty, a hard constraint on the failing architecture, and a different architecture that sidesteps the failure mode by construction – is itself informative about where the real leverage lies. The clearest lesson is that a method's reliability can depend more on what structure is built into it than on how it is trained: the HNN's advantage here is not that it was tuned better, but that it was set up so that a known conservation law could not be violated regardless of training quality.

We tested this advantage against the actual system that motivated the paper: the full periodically δ-kicked chaotic double well, not just its smooth sub-flow. The result is genuinely mixed, and we report it as such rather than rounding it to a clean verdict. The Hamiltonian neural network correctly reproduces the qualitative chaotic folding – the horseshoe bend and subsequent lobe-splitting visible in the exact solution – through four kick periods, a real success on the harder test this paper set out to attempt. At the same time, its quantitative advantage erodes steadily and predictably: pointwise error grows roughly 15× over those four periods, consistent with chaotic amplification of small errors in the learned dynamics, and by period 3–4 has caught back up to the error level of the original failing configuration this paper began by diagnosing. Both halves of that sentence are the finding. A structurally-informed deep learning approach can succeed where a naive one fails, by a wide and practically meaningful margin, on a real chaotic system – but chaos does not forgive even small residual errors indefinitely, and we did not test far enough in kick-period count to learn whether this specific approach's advantage stabilizes, degrades further, or is eventually overtaken entirely. That longer-horizon test, together with the higher-dimensional regimes where an SOT alternative would matter most, are the necessary next steps before any general claim about a PINN-family alternative to SOT can be assessed.

ACKNOWLEDGMENTS

[Acknowledgments – fill in.]

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